LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

DEPARTMENT OF ECONOMICS

Complexity Adjusted Comparisons of LPs and VARs: Evidence from Monte-Carlo Simulations

Aaron Liebig and Patrick Tenner

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Supervisor: Prof. Dr. Derya Uysal

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1. Introduction (1.5-2 pages)

To infer statements about the dynamic response to exogeneous shocks, the estimation of the corresponding impulse response has relied on two fundamental approaches, either through vector autoregressive (VAR) or Local Projection (LP) models. Since the introduction of the latter, an extensive literature has emerged discussing the relative merits of the two. A first strand of literature has established that, under relatively general conditions, LPs and VARs estimate the same population impulse responses (Plagborg-Moller & Wolf, 2021; Fusari et al. 2026). How-ever, in the discussion of finite sample behavior, LPs were often viewed as less sensitive to dy-namic misspecification whereas VARs were seen as potentially more statistically efficient (Li et al., 2024; Montiel Olea et al., 2025). Ludwig (2026) offers an analytical approach of explaining the observed finite sample bias-variance trade-off. He poses that LPs can be represented as a sequence of VAR estimations in-creasing in lag length as the estimated horizon increases and thus argues for a renewed thinking about how we can compare LPs and VARs – not by lag length, but by effective complexity.

The first part of this paper aims to build on the discussion outlined above by asking whether conventional finite-sample conclusions are robust to alternative benchmark comparisons that relax the standard equal-lag restriction. To answer this question, the paper operationalizes Lud-wig's (2026) central implication through a complexity-adjusted benchmarking exercise in which a fixed LP estimator is compared to progressively higher-order VAR specifications that proxy increasing effective dynamic flexibility (De Graeve & Westermark, 2025). This comparison is conducted across empirically relevant sample sizes (see e.g. Herbst & Johannsen 2021) to as-sess how the resulting bias–variance trade-off depends on the available data.

The central research question is: Do LPs retain their apparent robustness advantage once they are compared not only to equal-lag VARs, but also to higher-order VARs that approximate the effective complexity implied by LP estimation? The contribution is not to establish a new LP–VAR equivalence result, but to ask whether standard Monte Carlo comparisons between LPs and equal-lag VARs are affected by the fact that LPs correspond to a more flexible dynamic benchmark.

The second part of the paper introduces controlled dynamic misspecification. While the baseline experiment studies complexity-adjusted LP–VAR comparisons under correctly specified linear dynamics, the main practical argument for LPs concerns their robustness when the estimated linear model is only an approximation to the true DGP (Montiel Olea et al., 2021). To study this channel explicitly, the paper introduces a controlled degree of misspecification into the DGP and examines how the relative performance of LPs, equal-lag VARs, and higher-order VARs chang-es as misspecification becomes more severe.

The simulation design proceeds in three steps. First, we consider a correctly specified bivariate VAR(4) DGP, allowing LP(4), VAR(4), and higher-order VARs to be compared under common autoregressive information. Second, we introduce dynamic misspecification by adding a mov-ing-average shock component, thereby generating VARMA(4,1) dynamics while continuing to estimate finite-order VARs and LPs. Third, we introduce nonlinear misspecification through quadratic terms in the DGP while retaining linear estimators. This structure separates the role of dynamic approximation error from nonlinear functional-form misspecification and allows us to study whether conventional LP–VAR finite-sample rankings are robust to complexity-adjusted VAR benchmarks.

For each estimator, horizon, DGP, and sample size, we compute bias, variance, mean squared error, and empirical coverage rates. Bias and variance describe the finite-sample trade-off in point estimation, while coverage assesses whether the corresponding inference procedures re-main reliable under increasing dynamic or nonlinear misspecification.

2. Theoretical Background (total: 5-6.5 pages)

Impulse response functions can be nonparametically defined as the difference between two conditional forecasts of the outcome variable (e.g. Koop (1996)). Both forecasts are conditioned on the same history up to period $t-1$, but differ in the realization at time t. While one conditions on a shock of size δ to the variable of interest, the other one does not. Following Ramey (2016) shocks hereby refers to a significant exogenous force representing unanticipated movements in exogeneous variables, that is uncorrelated with other current and lagged control variables and other exogeneous shocks. The observed difference in the projected paths at horizon h then identifies the causal effect of the shock in population:

$$\mathrm{IR}(h, \delta) \equiv \mathbb{E}\left[y_{t+h}|s_t = s_0 + \delta; \mathbf{x_t}\right] - \mathbb{E}\left[y_{t+h}|s_t = s_0; \mathbf{x_t}\right] \tag{1}$$

Vector Autoregressions and Local Projections serve as two complementary approaches to estimating impulse responses in finite samples. The former, introduced by Sims (1980), parametrizes the joint dynamics of all variables in a single autoregressive system and impulse responses are then obtained as iterated forecasts from the estimated companion matrix. The latter, established by Jordà (2005), takes a direct forecasting approach by estimating the responses at each horizon independently via OLS, projecting the future outcome directly onto current and lagged values of the data. The two approaches compound estimation error differently across horizons and, at equal lag order, occupy opposite ends of a bias-variance spectrum in finite samples (Li et al., 2024; Olea et al., 2025). Local projections tend toward higher variance, vector autoregressions toward higher bias under misspecification. More recently, however, Ludwig (2026) shows that the equal-order comparison is itself misleading. At a given lag order p, LP(p) is strictly more general than VAR(p), so the conventional bias-variance trade-off conflates the choice of estimator with the choice of model complexity.

2.1 Introduction to VARS (1.5 - 2 pages)

Following Lüthkepohl (2006), we starting by defining a zero mean VAR(p) model of the form where the outcome of interest, y_t, is a $Kx1$ random vector, A_i are fixed KxK coefficient matrices and u_t are a K dimensional white noise processes with $\mathbb{E}\left[u_t\right] = 0$, $\mathbb{E}\left[u_t u_t'\right] = \sum_u$, i.e. being a nonsingular covariance matrix, and $\mathbb{E}\left[u_t u_s\right] = 0$ for $t \neq s$.

$$y_t = \nu + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \quad t = 0, \pm 1, \pm 2, \ldots \tag{2}$$

or simply

$$Y_t = \boldsymbol{\nu} + \mathbf{A} Y_{t-1} + U_t \tag{3}$$

where

$$Y_t := \begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{bmatrix}, \boldsymbol{\nu} := \begin{bmatrix} \nu \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \mathbf{A} := \begin{bmatrix} A_1 & A_2 & \cdots & A_{p-1} & A_p \\ I_K & 0 & \cdots & 0 & 0 \\ 0 & I_K & & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & I_K & 0 \end{bmatrix}, U_t := \begin{bmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{(Kp\times 1)}$$

Assuming Y_t satisfies the stability condition such that all roots, z, of the characteristic equation lie strictly outside the complex unit circle, the process is covariance stationary and admits a Wold moving-average representation.

$$det(I_{Kp} - \mathbb{A}z) \neq 0 \text{ for } z \leq 1 \leftrightarrow \tag{4}$$

2.2 Introduction to Local Projections (1.5 -2 pages)

Whereas the vector autoregression recovers the impulse response function by first estimating the full system dynamics and then interating the fitted model forward, the local projection approach estimates the same object directly, horizon by horizon, through a sequence of OLS regressions.

Assuming orthogonality of the shock with the projection error term and a linear projection model, we can obtain the impulse response to the shock at time h using the following OLS model:

$$y_{t+h} = \tag{5}$$

2.3 On the Equivalence of VARs and Local Projections (2-2.5 pages)

As Local Projection inference gained in popularity over the last couple of years, more and more scholars have started comparing the asymptotical and finite sample trade-offs of both methods.

Here I will discuss the theoretical results that establish the equivalence of VARs and LPs under certain conditions, as shown by Plagborg-Moller & Wolf (2021) and Fusari et al. (2026). I will also explain how this equivalence is affected by finite sample properties and dynamic misspecification, as discussed by Li et al. (2024), Montiel Olea et al. (2025), and Ludwig (2026).

The conventional finite-sample comparison between local projections and vector autoregressions is typically conducted at equal lag order, pitting LP(p) against VAR(p). Ludwig (2026) shows that this convention rests on a misleading notion of parsimony. His Theorem 1 establishes an exact finite-sample identity: the h-step prediction of an LP(p) is the forward iteration of a sequence of one-step VAR predictions whose lag order rises from p to $p + h - 1$. An equal-order VAR(p) is therefore a restricted special case of LP(p), obtained by holding the lag order fixed along the recursion rather than allowing it to increase. The familiar empirical pattern—lower approximation bias but higher sampling variance for LP relative to a same-order VAR—is, from this vantage point, largely a mechanical consequence of comparing a more general specification with its own restriction, rather than evidence of an intrinsic property of either estimator. This matters because it leaves the standard reading of the bias-variance trade-off, as articulated by Plagborg-Møller and Wolf (2021) and Montiel Olea et al. (2025), confounded: at equal lag order, the method effect (LP versus VAR) cannot be separated from the complexity effect (more versus fewer effective autoregressive parameters).

3. Baseline Simulation (4-5 pages)

This confounding motivates the first part of our study. Because Ludwig's contribution is analytical and deliberately abstains from a simulation-based horse race, the question of whether the conventional finite-sample conclusions survive once the equal-lag restriction is relaxed remains open. We therefore operationalize Ludwig's central implication through a complexity-adjusted benchmarking exercise that holds parsimony more nearly fixed, comparing a fixed LP(4) estimator against progressively higher-order VAR specifications that proxy its increasing effective dynamic flexibility. Crucially, because the identity in Theorem 1 maps LP(p) to a sequence of VARs rather than to any single higher-order VAR, no individual VAR is the appropriate counterpart; instead, LP(4) is read as a reference line against the full VAR order frontier, against which its bias, variance, and coverage can be located. We frame the exercise as a test of robustness rather than as a foregone conclusion: the aim is to assess whether the conventional bias-variance verdict persists, dissolves, or reverses once the comparison is placed on more comparable terms, and to identify the features of the data-generating process under which each procedure is favored.

3.1 Simulation Design

Let $y_t = (y_{1t}, y_{2t})'$ be a bivariate time series generated by the following VAR(4) process:

$$y_t = \sum_{l=1}^{4} A_l y_{t-l} + u_t, \quad u_t = B\epsilon_t \tag{6}$$

3.2 Simulation Results

4. Extension 1: Dynamic Misspecification (3-4 pages)

4.1 Simulation Design

4.2 Simulation Results

5. Extension 2: Functional Misspecification (3-4 pages)

5.1 Simulation Design

5.2 Simulation Results

6. Discussion (1.5-2 pages)

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